# Discrete Isogonal Nets with Similar Parallelograms


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|---|---|---|---|
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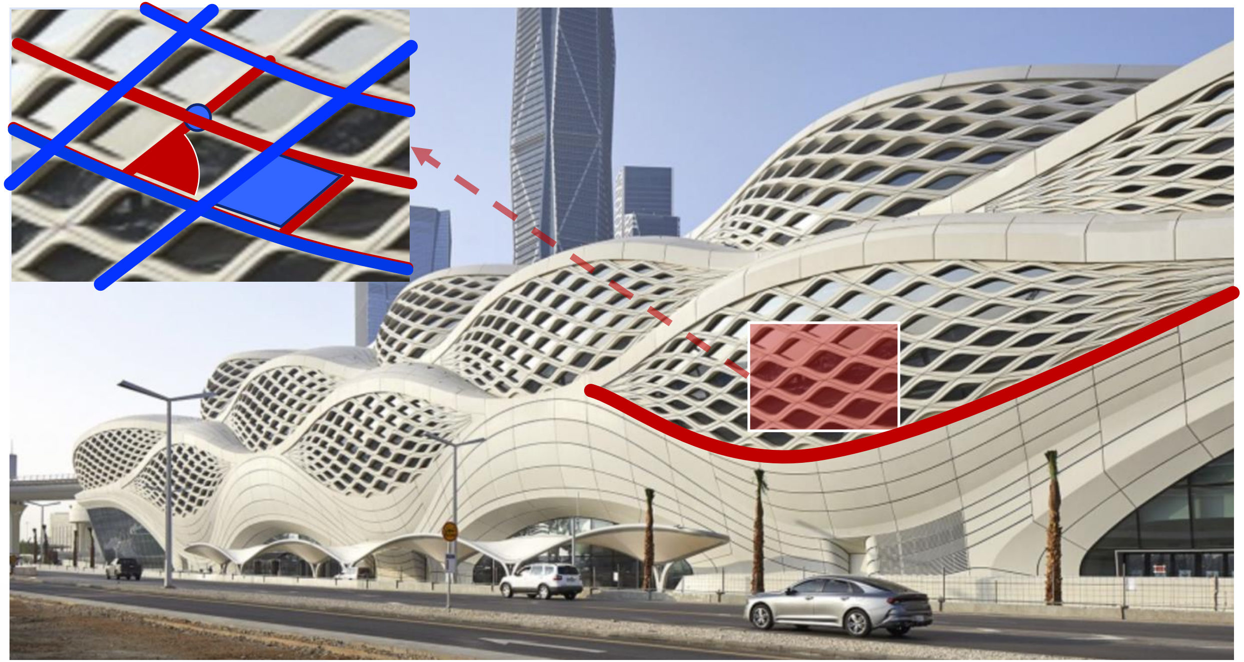
# Background

- High-quality surfaces are essential in architecture and product design



KAFD Metro Station in Riyadh, designed by ZHA

- regular vertex
- smooth curve
- equal opposite sides
- right angle
- flat face
- tessellation density
- feature line alignment

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# Background

## Orthogonal nets (O-nets)

- basic for other specialized quad nets,
  e.g. principal curvature net, minimal net,…
- offer aesthetic appeal and engineering practicality

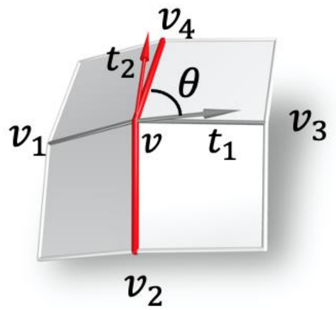
## Isogonal nets (I-nets)

- Constant curve angle beyond 90°
- Compelling alternative to O-nets
- Broader control over shape
- Unique visual effects
- Adaptable structural layouts

Yas Hotel, Abu Dhabi

Phoenix Media Center, Beijing

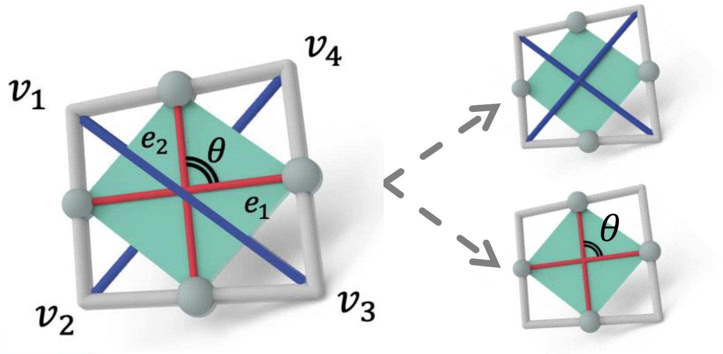
# Overview of I-nets




O-net: $\theta = \dfrac{\pi}{2} \Leftrightarrow (e_3 - e_1)\cdot(e_4 - e_2) = 0$ ⇠ $e_i = \dfrac{v_i - v}{||v_i - v||}, (i = 1,2,3,4)$

I-net: $\theta = const. \Leftrightarrow t_1 \cdot t_2 = \cos\theta$ ⇠ $t_1 = \dfrac{e_3 - e_1}{||e_3 - e_1||}, t_2 = \dfrac{e_4 - e_2}{||e_4 - e_2||}$

Defined unit tangents - - -> increase computation cost




Mid-edge subdivided quad is a parallelogram ▱

O-net: $\theta = \dfrac{\pi}{2} \Leftrightarrow (v_3 - v_1)^2 = (v_4 - v_2)^2$ ◊

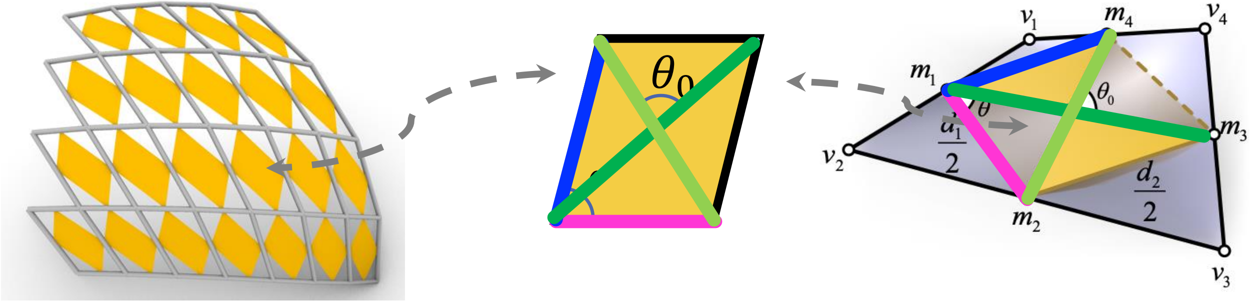
I-net: $\theta = const.$
$\Leftrightarrow$ ?

Checkerboard pattern (CBP) - - - - -> How to represent I-net by edge lengths?

Felix Dellinger, Xinye Li, Hui Wang#. *Discrete orthogonal structures*. Computers & Graphics 2023.

# Proposed Method

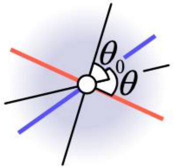
- I-nets using similar mid-edge subdivided parallelograms




$$\frac{||d_2||}{||d_1||} = \lambda$$

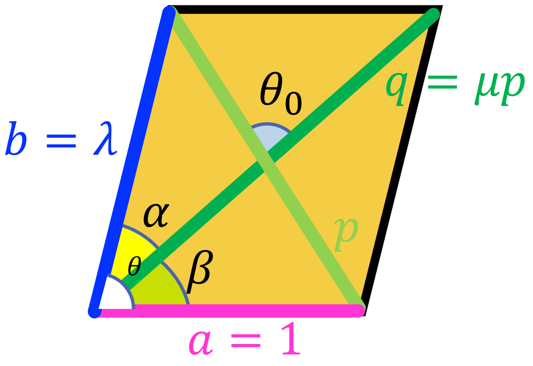
$$\frac{||m_1 m_3||}{||m_2 m_4||} = \mu$$

- Constraining edge ratios instead of angles
- Generalizations of O-nets
  - Rhombus
  - Rectangle
  - Square




- Generalization to isogonal 4-webs (I-webs)
- Versatility of I-nets and I-webs in high-quality surface

# Parallelogram Geometry

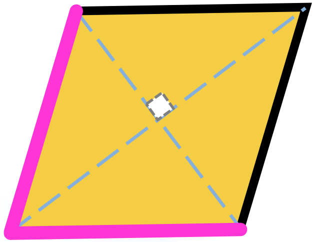



$$a:b:p:q=1:\lambda:\sqrt{\frac{2(1+\lambda^2)}{1+\mu^2}}:\mu\sqrt{\frac{2(1+\lambda^2)}{1+\mu^2}}$$

$$\cos\theta=\frac{(1+\lambda^2)(\mu^2-1)}{2\lambda(1+\mu^2)},\cos\theta_0=\frac{(1+\mu^2)(\lambda^2-1)}{2\mu(1+\lambda^2)}$$

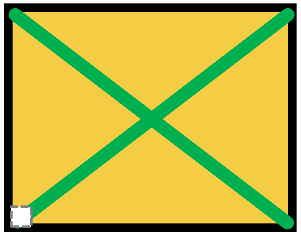
$$\cos\alpha=\frac{3\lambda^2\mu^2+\lambda^2+\mu^2-1}{2\lambda\mu\sqrt{2(1+\lambda^2)(1+\mu^2)}},\cos\beta=\frac{\lambda^2\mu^2-\lambda^2+3\mu^2+1}{2\mu\sqrt{2(1+\lambda^2)(1+\mu^2)}}$$

$\lambda=1 \Leftrightarrow \theta_0=\frac{\pi}{2}$



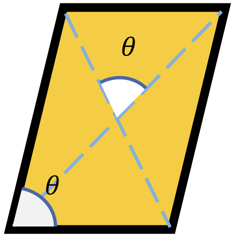
Rhombu
s

$\mu=1 \Leftrightarrow \theta=\frac{\pi}{2}$



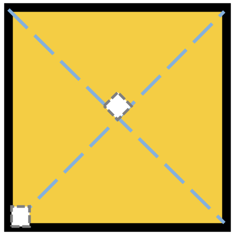
Rectangl
e

$\lambda=\mu \Leftrightarrow \theta=\theta_0$




$\theta$-Parallelogram

$\lambda=\mu=1 \Leftrightarrow \theta=\theta_0=\frac{\pi}{2}$



Squar
e

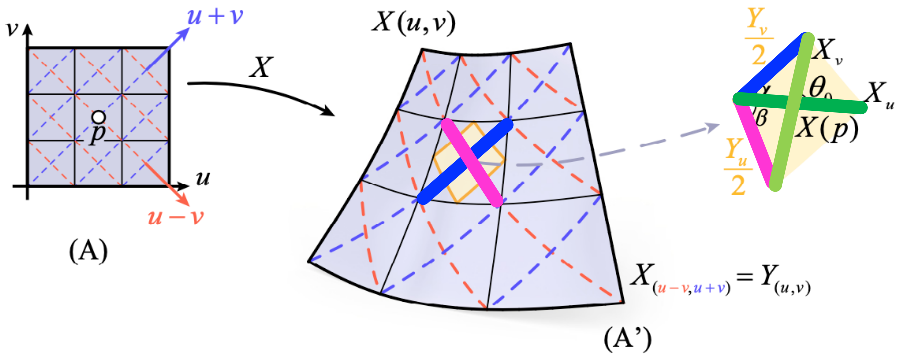
# Isogonal Nets and Webs

Surface parametrization:

$$X: (u, v) \subset \mathbb{R}^2 \rightarrow X(u, v) \subset \mathbb{R}^3$$

Diagonal parametrization:

$$Y(u, v) = X(u - v, u + v)$$




constant $\frac{||X_u||}{||X_v||} := \mu$

constant $\frac{||Y_v||}{||Y_u||} := \lambda$

$$\Longrightarrow \quad Y_v^2 = \lambda^2\, Y_u^2 \Leftrightarrow (X_u + X_v)^2 = \lambda^2\, (X_u - X_v)^2$$

$$\Downarrow$$

$$\theta(\lambda, \mu), \theta_0(\lambda, \mu), \alpha(\lambda, \mu), \beta(\lambda, \mu)$$

$$\cos\theta = \frac{(1+\lambda^2)(\mu^2-1)}{2\lambda(1+\mu^2)},$$

$$\cos\theta_0 = \frac{(1+\mu^2)(\lambda^2-1)}{2\mu(1+\lambda^2)}$$

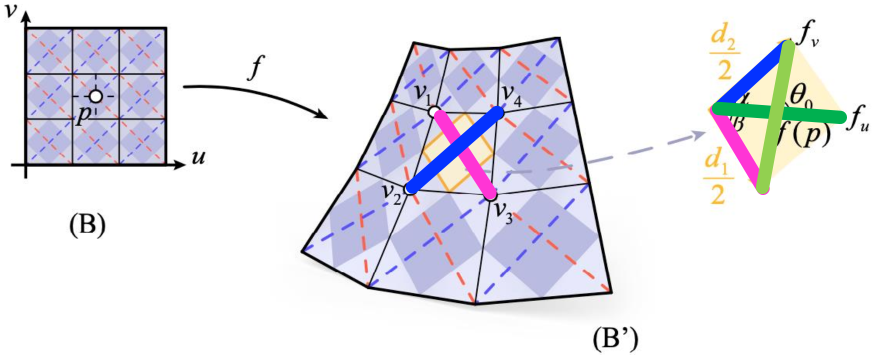
$$\cos\alpha = \frac{3\lambda^2\mu^2+\lambda^2+\mu^2-1}{2\lambda\mu\sqrt{2(1+\lambda^2)(1+\mu^2)}},$$

$$\cos\beta = \frac{\lambda^2\mu^2-\lambda^2+3\mu^2+1}{2\mu\sqrt{2(1+\lambda^2)(1+\mu^2)}}$$

**Theorem**

If the ratios $\frac{||X_u||}{||X_v||}$ and $\frac{||Y_v||}{||Y_u||}$ are constant, $X(u,v)$ is **isogonal (I-net)**, and $X(u,v)$ and $Y(u,v)$ form an **isogonal 4-web (I-web)**.

# Discrete Isogonal Nets and Webs




Quad mesh $f: \mathbb{Z}^2 \to \mathbb{R}^3$ :

vertices $v_i(1, \ldots, 4)$ in each quad face

Discrete first-order smooth derivatives :

$$d_1 = v_3 - v_1,\ d_2 = v_4 - v_2$$

Discrete second-order smooth derivatives :

$$f_u = \frac{1}{2}(v_3 + v_4 - v_1 - v_2),\ f_v = \frac{1}{2}(v_1 + v_4 - v_2 - v_3)$$

constants $\frac{||d_2||}{||d_1||} := \lambda,\ \frac{||f_u||}{||f_v||} := \mu$ $\Longrightarrow$ $\theta(\lambda,\mu), \theta_0(\lambda,\mu), \alpha(\lambda,\mu), \beta(\lambda,\mu)$

**Theorem**

If the diagonal ratio $\frac{||d_2||}{||d_1||}$ and medial-line ratio $\frac{||f_u||}{||f_v||}$ are constant, $f$ is **isogonal (DI-net)**, and $f$ and its diagonal quad mesh $d$ form a discrete **isogonal 4-web (DI-web)**.

# Special cases

- $\lambda = 1 \iff \theta_0 = \frac{\pi}{2} \iff X_u \perp X_v$

  **Definition.** An I-net is an orthogonal net (**O-net**) if $\lambda$=1.

  **Definition.** A DI-net is orthogonal (**DO-net**) if $\lambda$=1.

  **Corollary.** A DO-net consists of similar rhombuses in the CBP.

- $\mu = 1 \iff \theta = \frac{\pi}{2} \iff Y_u \perp Y_v$

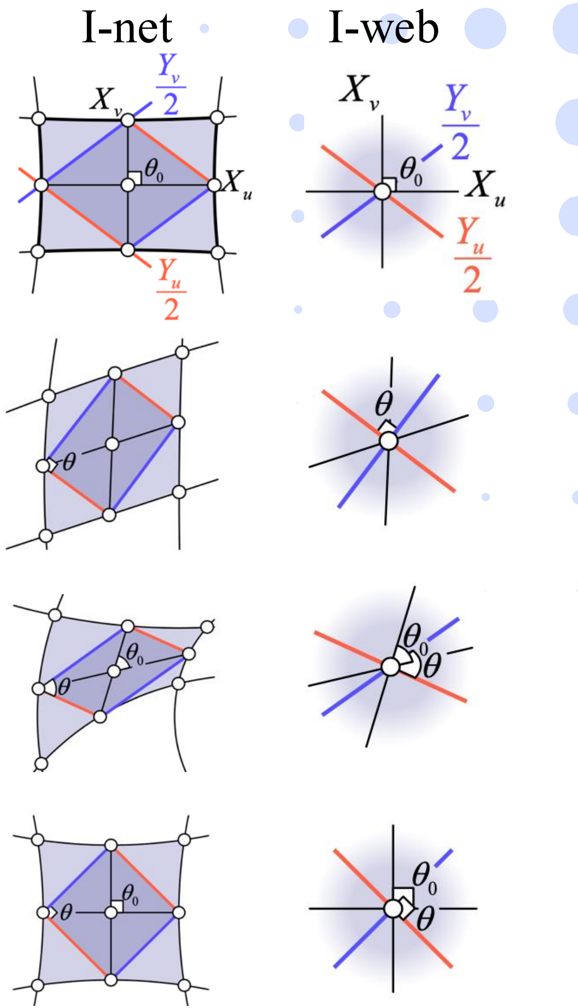
  **Corollary.** A DI-net with $\mu$=1 consists of similar rectangles in the CBP.

- $\lambda = \mu \iff \theta = \theta_0 \iff \frac{||X_u||}{||X_u||} = \frac{||X_u+X_v||}{||X_u-X_v||}$

  **Corollary.** A DI-net with $\theta$-Quads consists of similar θ-Parallelograms in the CBP.

- $\lambda = \mu = 1 \iff \theta = \theta_0 = \frac{\pi}{2} \iff X_u \perp X_v,\ Y_u \perp Y_v$

  **Corollary.** A DO-net with $\theta$-Quads consists of squares in the CBP.

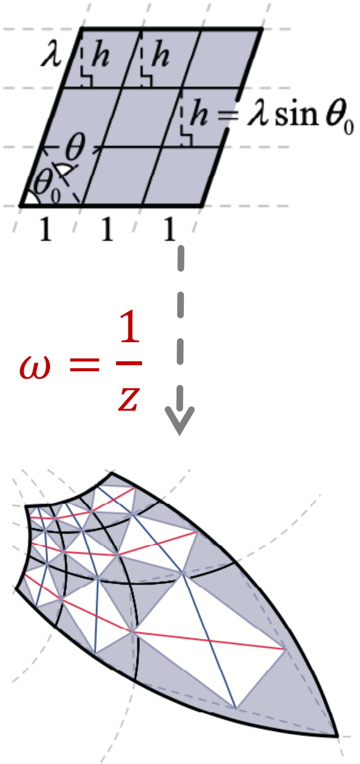


# Construction of I-nets

- I-nets on planar sheared grid
- I-nets on developable surfaces
- I-nets from isothermal parametrizations
- I-nets by computational design

# Construction of I-nets

- I-nets on planar sheared grid




$$C_1 = \{z = t + i\mathcal{A}, \frac{\mathcal{A}}{h} \subseteq \mathbb{Z}, 0 \notin \mathcal{A}, h \in \mathbb{R}\}$$

$$C_2 = \{z = t + \mathcal{B} + i(kt), k = \tan\theta \in \mathbb{R}, \mathcal{B} \subseteq \mathbb{Z}, 0 \notin \mathcal{B}\}$$

$$\omega = \frac{1}{z}$$

$$u + iv = \frac{t}{t^2+\mathcal{A}^2} - i\frac{\mathcal{A}}{t^2+\mathcal{A}^2}$$

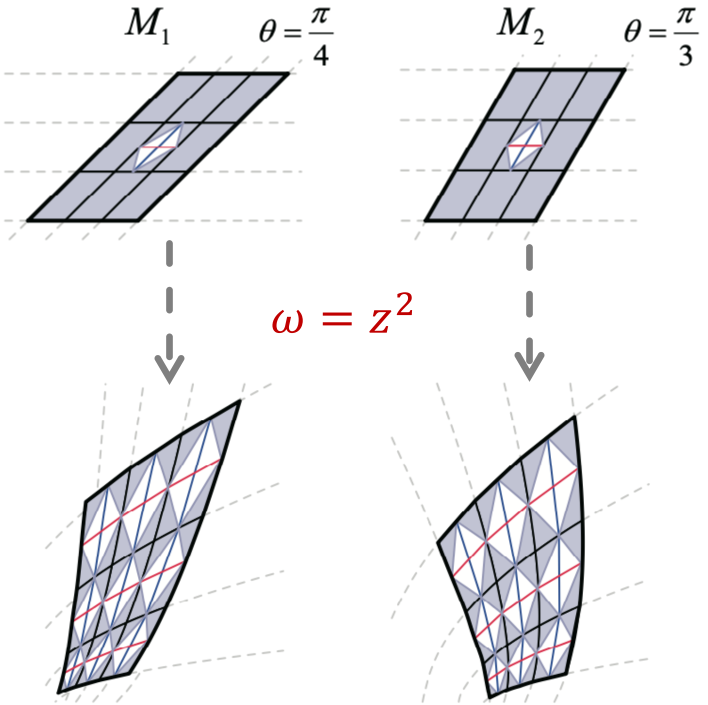
$$u + iv = \frac{t+\mathcal{B}}{(1+k^2)t^2+2\mathcal{B}t+\mathcal{B}^2} - i\frac{kt}{(1+k^2)t^2+2\mathcal{B}t+\mathcal{B}^2}$$

$$\Downarrow$$

$$u^2 + (v + \frac{1}{2\mathcal{A}})^2 = \frac{1}{4\mathcal{A}^2}, \ (u - \frac{1}{2\mathcal{B}})^2 + (v - \frac{1}{2k\mathcal{B}})^2 = \frac{1 + k^2}{4k^2\mathcal{B}^2}$$

# Construction of I-nets

- I-nets on planar sheared grid

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$$C_2 = \{z = t + \mathcal{B} + i(kt), k = \tan\theta \in \mathbb{R}, \mathcal{B} \subseteq \mathbb{Z}, 0 \notin \mathcal{B}\}$$

$$\downarrow \omega = z^2$$

$$u + iv = t^2 - \mathcal{A}^2 + i(2\mathcal{A}t)$$

$$u + iv = (1 - k^2)t^2 + 2\mathcal{B}t + \mathcal{B}^2 + i(2kt^2 + 2k\mathcal{B}t)$$

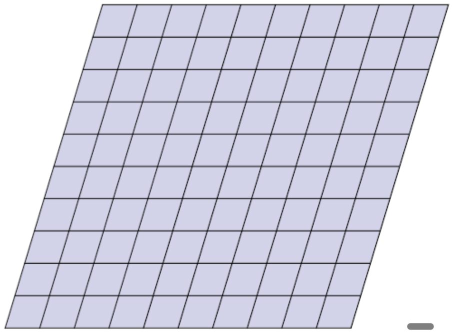
$$\Downarrow$$

$$v^2 - 4\mathcal{A}^2 u - 4\mathcal{A}^4 = 0$$

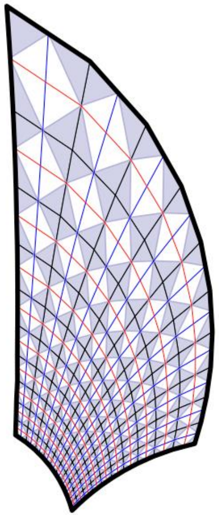
$$4k^2u^2 + (k^2 - 1)^2v^2 + 4k(k^2 - 1)uv + 4k^2\mathcal{B}^2(k^2 - 1)u - 8k^3\mathcal{B}^2 v - 4k^4\mathcal{B}^4 = 0.$$

# Construction of I-nets

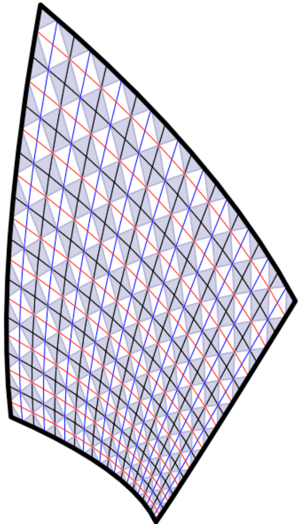
- I-nets on planar sheared grid



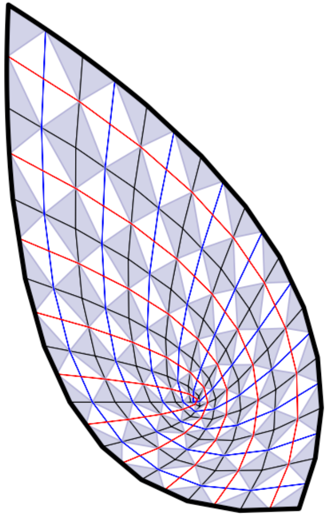
10 x 10 grid



$\omega = \frac{1}{z}$



$\omega = z^2$

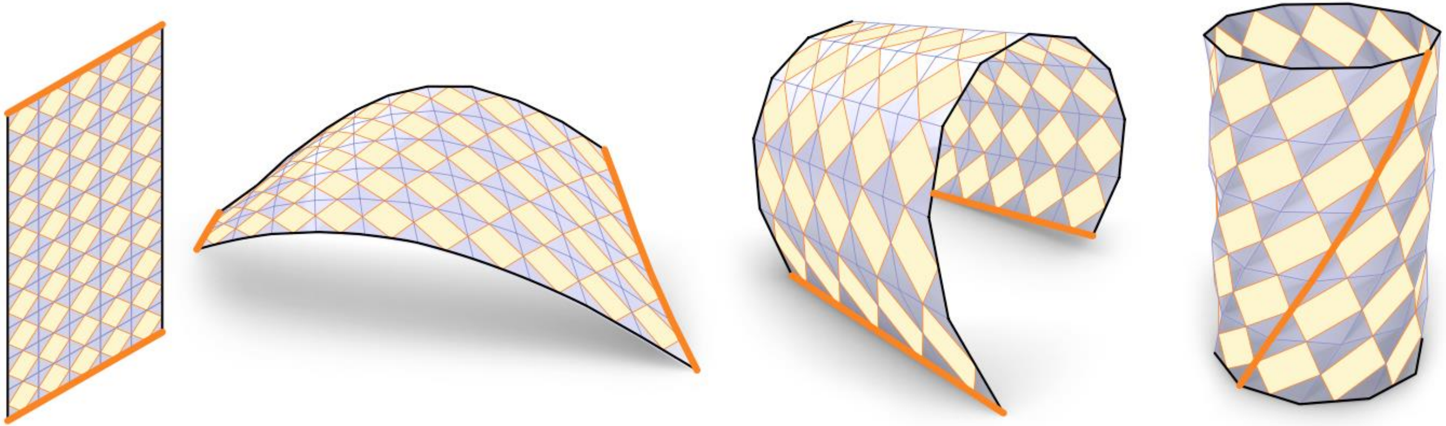


$\omega = z^2$

(Origin lies within the pre-image)

# Construction of I-nets

- I-nets on developable surfaces



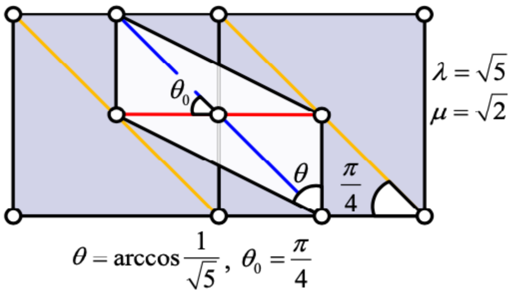
developable I-nets through isometric deformations

# Construction of I-nets

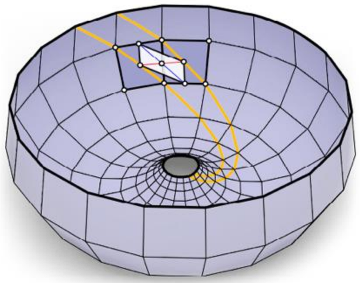
- I-nets from isothermal parametrizations

Isothermal parametrization $S(u,v)$:

- $F = S_u \cdot S_v = 0, \quad E = S_u^2 = G = S_v^2$
- Conformal
- Maps small squares from domain to small squares on the surface




$(\theta, \theta_0) = (arccos\frac{1}{\sqrt{5}}, 45°)$



# Construction of I-nets

- I-nets from isothermal parametrizations

Surface of revolution: $S(u,v) = f(v)\cos u, f(v)\sin u, g(v))$ $\Longrightarrow$ $F = 0$

$$\boxed{E = f^2(v) = f'^2(v) + g'^2(v) = G}$$

Isothermal parametrizations

- Cylinder: $f(v) = r, g(v) = r\,v$
- Sphere: $f(v) = r\,\mathrm{sech} v, g(v) = r\tanh v$
- Catenoid ($C$): $f(v) = r\cosh v, g(v) = r\,v$

- Helicoid ($H$): $S(u,v) = (r\sinh v\; cos u, r\sinh v\; sin\, u, ru)$
- Conjugate minimal surface: $(\cos t)\, C + (\sin t)\, H$, $t \in \mathbb{R}$

**Isothermal net:**

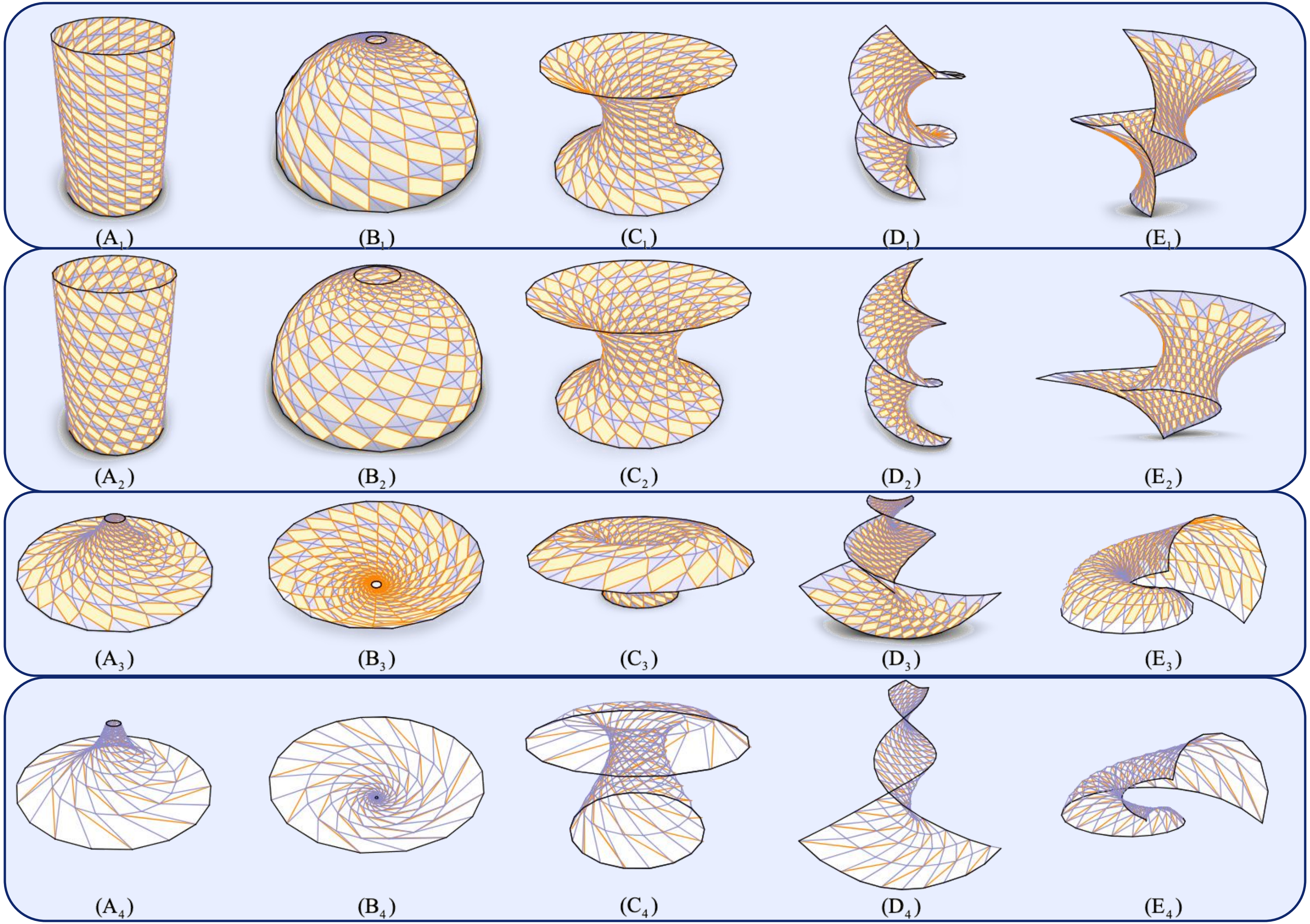
$(\theta, \theta_0) = (arccos\frac{1}{\sqrt{5}}, 45°)$

$(\theta, \theta_0) = (80°, 45°)$ :

**Möbius transf. :**

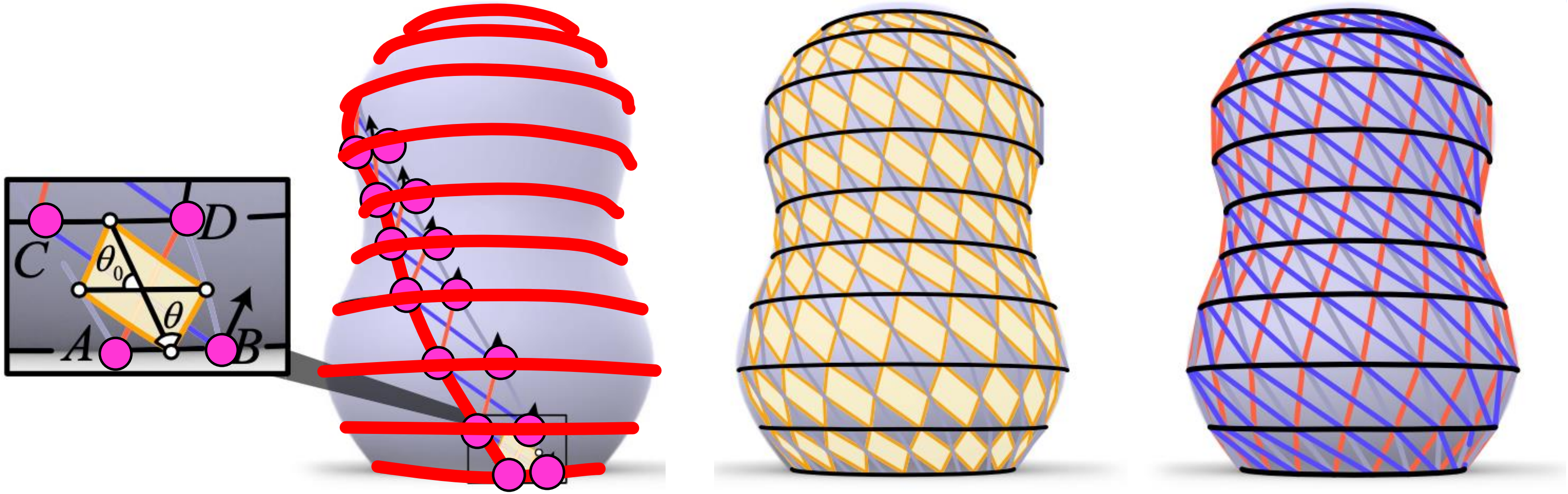
$(\theta, \theta_0) = (arccos\frac{1}{\sqrt{5}}, 45°)$

$(\theta, \theta_0) = (50°, 50°)$:



# Construction of I-nets

- I-nets formed by loxodromes and parallels




(A) (B) (C)

# Computational Design

- Initialization of I-nets
- Optimization

$$E_{+} = E_{\square} + \omega_1 E_{*} + \omega_2 E_{fair} + \omega_3 E_{clos} + \omega_4 E_{app} + \omega_5 E_{bdry}$$

$$\frac{||v_4 - v_2||}{||v_3 - v_1||} = \lambda, \quad \frac{||v_3 + v_4 - v_1 - v_2||}{||v_1 + v_4 - v_2 - v_3||} = \mu$$

Assign const. $\bar{\lambda}$, $\bar{\mu}$ or $\bar{\theta}$, $\overline{\theta_0}$

$$E_{fair} = \sum_{i \in polyline} (2\mathbf{v}_i - \mathbf{v}_{il} - \mathbf{v}_{ir})^2$$

$$E_{clos} = \sum_{i \in vertices} (\mathbf{v}_i - \bar{\mathbf{v}}_i)^2$$

$$E_{app} = \sum_{i \in vertices} ((\mathbf{v}_i - \mathbf{p}_i) \cdot \mathbf{n}_i)^2$$

$$E_{bdry} = \sum_{i \in boundary} ((\mathbf{v}_i - \mathbf{p}_i) \times \mathbf{e}_{i1})^2$$

# ArchGeo — Geometry Processing Library

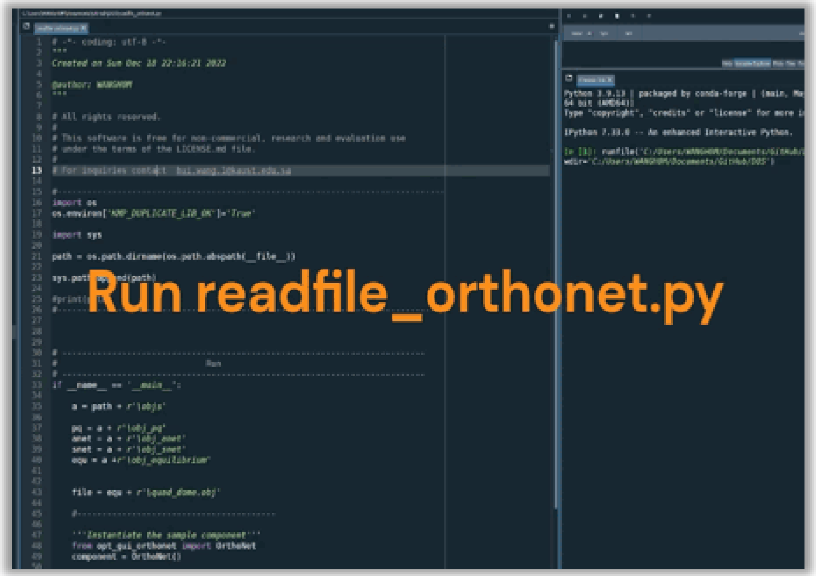
Github Page

## Geometric Visualization

Point cloud, Polylines, Bezier / B-spline / NURBS curves / surfaces, Meshes, Curvatures, Vector fields...

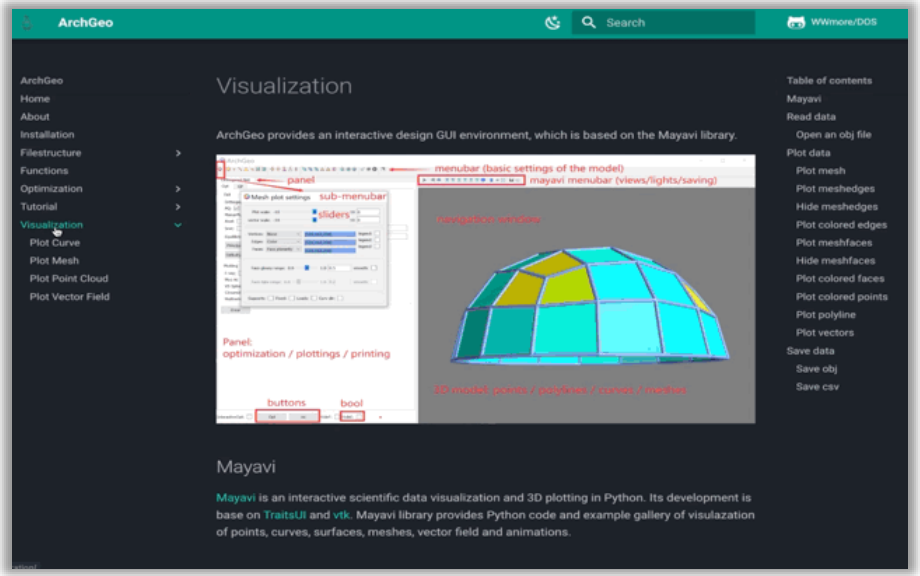
## Geometric Optimization

Curvature nets, Gridshell structures, Paneling, Parametrizations, Interactive editing...




ArchGeo Window

https://github.com/WWmore/DOS




ArchGeo Documentation Page

https://www.huiwang.me/mkdocs-archgeo/

Felix Dellinger, Xinye Li, Hui Wang#. *Discrete orthogonal structures*. Computers & Graphics 2023.

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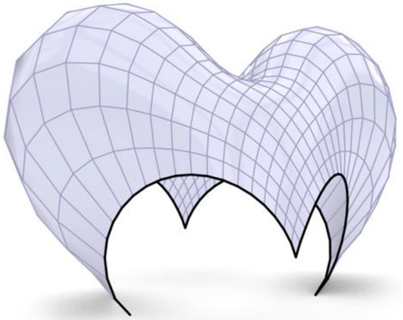
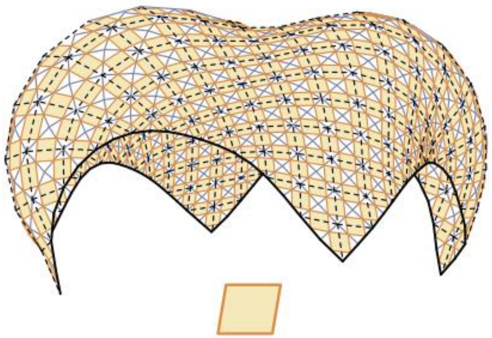
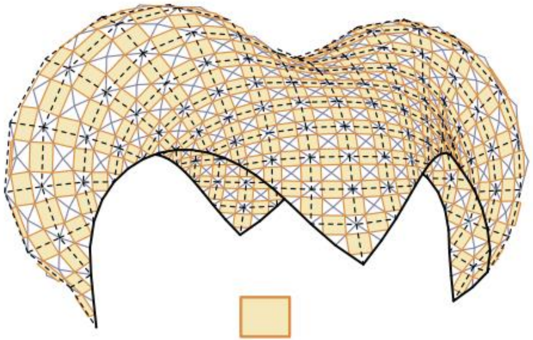
# Results



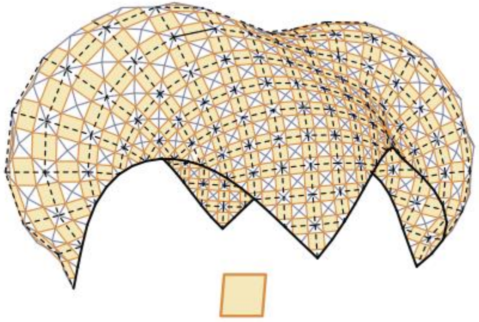
Initial mesh



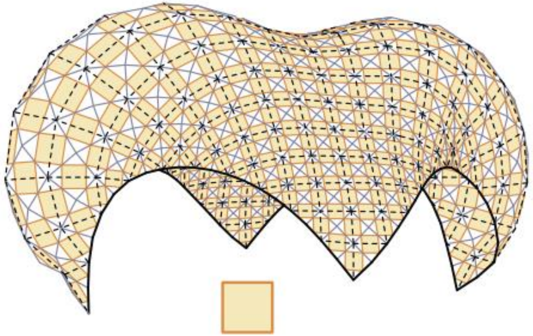
(A) $(\theta,\theta_0)=(80.42°,84.30°)$



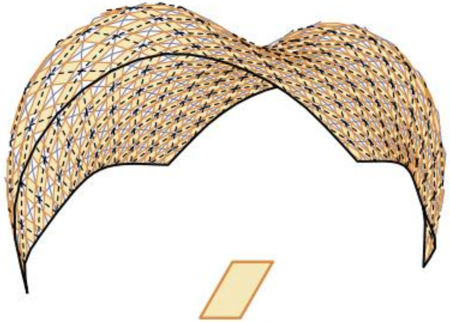
(B) $(\theta,\theta_0)=(90°,77.92°)$



(C) $(\theta,\theta_0)=(85.27°,90°)$

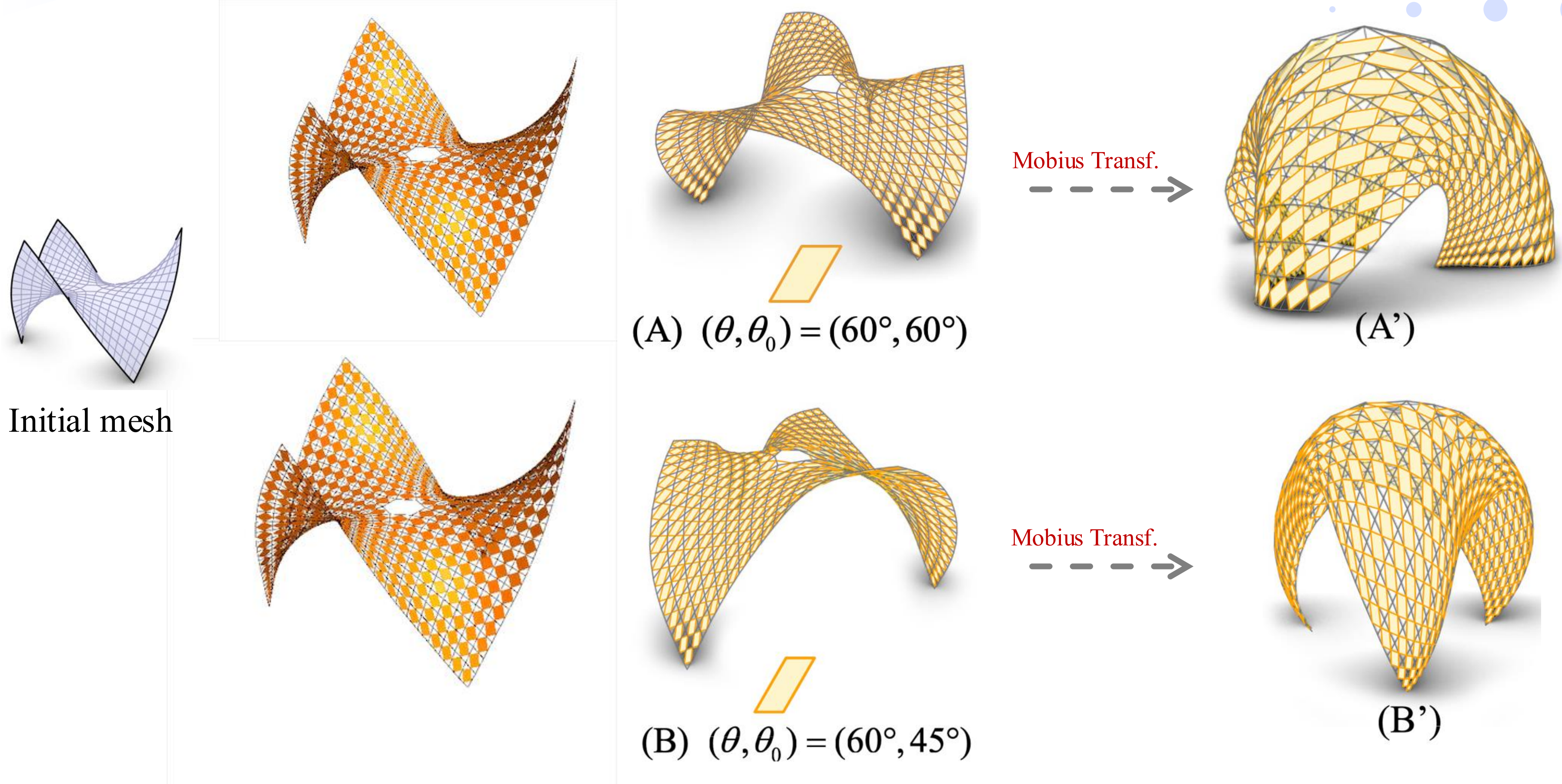


(D) $(\theta,\theta_0)=(90°,90°)$



(E) $(\theta,\theta_0)=(60°,60°)$

# Results



Initial mesh
Mobius Transf.
(A) $(\theta,\theta_0)=(60°,60°)$
(A')
Mobius Transf.
(B) $(\theta,\theta_0)=(60°,45°)$
(B')

# Conclusion

- A novel method for I-net and I-web using similar mid-edge subdivided parallelograms
- Expand the scope of geometric design possibilities and ensures compatibility with orthogonal nets
- Simplify the optimization process by constraining edge ratios instead of angles

# Future work

- Integrate with other specialized nets, e.g. Weingarten surfaces
- Potential applications:
  - panelling design for freeform architectural surfaces
  - gridshell structures with uniform knots
  - repetitive pattern design
  - auxetic quads with similar elements
  - periodic structures in additive manufacturing
  - woven fabrics with regular patterns
  - ……

# Thank you！

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SPM 2025
Hangzhou, China · OCT 29–NOV 2

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